

Tissues & the Integumentary System

Four tissue types build every organ you will study. The skin uses all four at once.

BY THE END OF THIS PAGE YOU SHOULD BE ABLE TO

- Name the four tissue types and say what each one is for
- Explain why epithelium comes in different shapes and layers
- Say what makes connective tissue different from everything else
- Name the three layers of skin and what lives in each
- Classify a burn by depth, and say why a painless burn is the serious one

1 Four tissues, and only four

Every organ in the body is built from these four. Learn the job, not the list.

TISSUE	BUILT FOR	WHERE YOU MEET IT
Epithelial	Covering, lining, releasing, absorbing	Skin surface, gut lining, glands, airways
Connective	Supporting, binding, carrying	Bone, cartilage, tendon, fat, and blood
Muscle	Making movement by shortening	Skeletal, cardiac, smooth
Nervous	Sensing and signaling fast	Brain, spinal cord, every nerve

Two facts about epithelium explain a lot later. It sits on a **basement membrane**. And it has no blood vessels of its own.

Instead it is fed by diffusion from the connective tissue underneath. That is why a scrape only bleeds once it goes deep enough to reach the tissue below.

THINK OF IT THIS WAY

Epithelium is wallpaper. It covers everything and takes the wear. It has no plumbing of its own, so it lives off the wall behind it.

2 Epithelium: shape and layers tell you the job

You can predict what a lining does by looking at how it is built.

- **One layer, flat cells.** Thin, so things cross easily. This is what you find in the air sacs of the lung and in capillaries.
- **One layer, tall cells.** Built to absorb and to release. This lines the gut.
- **Many layers.** Built to survive wear. This is the surface of the skin, the mouth, and the esophagus.

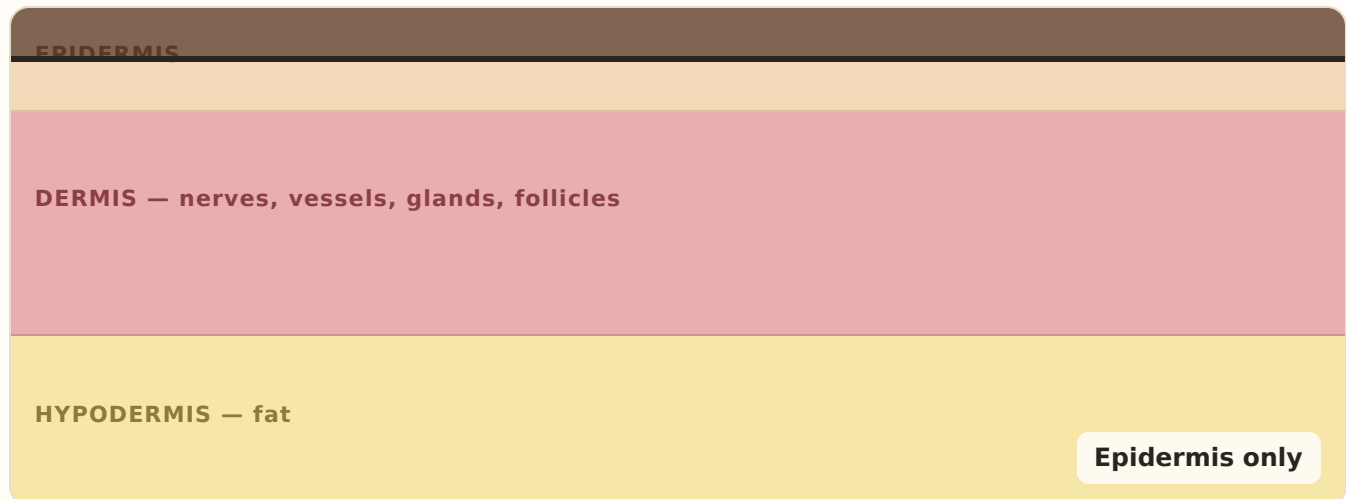
So the rule runs both ways. Thin means exchange. Thick means protection.

Connective tissue is the odd one out. Its cells are spread out in a **matrix** that they make themselves. Change the matrix and you change the tissue. Add mineral and you get bone. Use flexible protein and you get cartilage. Make the matrix a liquid and you get blood. That is why blood counts as connective tissue.

3 The skin, layer by layer

Drag down through the skin and watch what a wound reaches at each depth.

BURN DEPTH — DRAG DOWN THROUGH THE SKIN



Red, dry and painful — think sunburn. The dermis is untouched, so nerves, follicles and glands all survive. It heals itself in a few days with no scar, because the stem cells that rebuild epidermis are still there.

- **Epidermis.** Epithelium. No blood vessels. It replaces itself from the bottom up. It holds the pigment cells and the waterproofing keratin.
- **Dermis.** Connective tissue, and the working floor of the organ. Blood vessels, nerve endings, sweat glands, oil glands, and hair follicles all live here.
- **Hypodermis.** Mostly fat. It insulates, stores energy, and absorbs shock.

4 What the skin actually does

Skin is an organ, not a wrapper. It has five jobs worth knowing.

- 1 **Barrier.** It keeps germs out and water in. This is the main reason burns are dangerous.
- 2 **Temperature control.** Sweat glands lose heat. Vessels in the dermis widen or narrow to dump heat or hold it.
- 3 **Sensation.** Touch, pressure, temperature, and pain receptors sit in the dermis.
- 4 **Vitamin D.** Sunlight starts it in the epidermis. The liver and kidney finish it.
- 5 **Excretion.** A little salt and waste leave in sweat.

Notice how many of these are feedback loops from Module 1. Too warm, and the receptor is in the skin. The control center is the brain. The effectors are sweat glands and blood vessels *in the skin*.

5 Why a large burn is an emergency

The danger is not the wound itself. It is what the missing barrier lets happen.

- **Fluid loss.** Water evaporates straight out of the exposed tissue, and plasma leaks into the burned area. Blood volume drops, and shock follows.
- **Infection.** The barrier that kept germs out is gone, over a large area.
- **Heat loss.** Without the barrier, and without vessels narrowing, heat pours away.

WHY THE NURSE CARES

Fluid is calculated from how much body surface is burned, not from how much it hurts. The deepest burns often hurt the least, because the nerve endings in the dermis have been destroyed.

6 Check yourself

Five questions. The wrong answers explain themselves, so they are worth reading.

1. Why does a shallow scrape of the epidermis not bleed?

- a) The epidermis clots instantly
- b) The epidermis has no blood vessels of its own
- c) Keratin seals the wound first
- d) Blood vessels are only in the hypodermis

2. Blood is which type of tissue?

- a) Epithelial
- b) Connective
- c) Muscle
- d) Nervous

3. A burn destroys the epidermis and the whole dermis. What would you expect to find?

- a) Severe pain at the burn
- b) Little or no pain at the burn
- c) More sweating over the area
- d) Hair growing back quickly

4. The brain senses the body is too warm. Which skin structures are the effectors?

- a) The basement membrane
- b) Pigment cells in the epidermis
- c) Sweat glands and dermal blood vessels
- d) The keratin layer

5. Which lining would you expect in the air sacs of the lung, where gases must cross quickly?

- a) Many layers of flat cells
- b) One layer of flat cells
- c) One layer of tall cells
- d) Many layers of tall cells

Answer key

1. **The epidermis has no blood vessels of its own**

Right. Epithelium has no vessels. It is fed by diffusion from the dermis below.

2. **Connective**

Correct. Connective tissue is cells spread through a matrix, and here the matrix is liquid plasma.

3. **Little or no pain at the burn**

Correct. The nerve endings live in the dermis, and they were destroyed with it.

4. **Sweat glands and dermal blood vessels**

Right. Sweat loses heat by evaporating, and wide vessels bring heat to the surface.

5. **One layer of flat cells**

Correct. One flat layer is as thin as a lining gets, so gases cross easily.

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